L2L Framework: TSE Evaluation

Script 05 — RQ1 (Item Selection), RQ2 (Construct Recovery), RQ3 (Item Measurement Error)

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Overview

Three CFA/IRT model fits per fold define the evaluation:

- **Gold standard:** CFA/IRT fit to full fold (training + test, no masking).
- **Listwise deletion:** CFA/IRT fit to training data only.
- **Imputed:** CFA/IRT fit to training + imputed test.

Research Questions

- **RQ1 — Psychometric context utilization:** Select best LLM-tier on item-level metrics (weighted kappa, accuracy).
- **RQ2 — Construct-level recovery:** Compare gold, deletion, best L2L, best traditional on rank preservation (r) and scale fidelity (regression slope).
- **RQ3 — Item measurement error:** Quantify how imputation distorts IRT item parameters (discrimination, thresholds) relative to gold standard, using Biemer decomposition of item parameter MSE into systematic (Bias^2) and random (Variance) components. Counterfactual = listwise deletion.

Setup

```
library(lavaan)
library(mirt)
library(semTools)
library(viridis)
library(tictoc)

set.seed(721)

results_dir <- "D:/repos/UMD_classes_code/TSE_II_SURV721/results"
```

```r
llm_results  <- readRDS(file.path(results_dir,
                                  "results_item.rds"))
trad_results <- readRDS(file.path(results_dir,
                                  "results_trad_item.rds"))
load(file.path(results_dir, "fold_data.RData"))
load(file.path(results_dir, "cfa_reporting.RData"))
load(file.path(results_dir, "irt_models.RData"))
masking_tbl <- readRDS(file.path(results_dir,
                                 "masking_table.rds"))

#   tidyverse loaded last
library(tidyverse)
```

Data Preparation

```r
shared_cols <- c("survey", "fold_id", "n_masked", "tier",
                 "method", "row_id", "item", "actual",
                 "predicted", "cfa_baseline", "irt_baseline")

all_results <- bind_rows(
  trad_results |>
    filter(!is.na(predicted)) |>
    select(all_of(shared_cols)) |>
    mutate(method_type = "Traditional"),
  llm_results |>
    filter(!is.na(predicted)) |>
    select(all_of(shared_cols)) |>
    mutate(method_type = "LLM")
) |>
  filter(method != "gemma3_27b") |>
  mutate(
    method_label = recode(method,
      llama4_maverick = "Maverick",
      claude_sonnet   = "Claude Sonnet",
      mixgb           = "mixgb",
      miceRanger      = "miceRanger"),
    method_type = factor(method_type,
                         levels = c("Traditional", "LLM")),
    tier = factor(tier, levels = c("S", "P1", "P2")),
    survey_label = if_else(survey == "lapop",
                           "LAPOP Mexico",
                           "Latinobarómetro Colombia")
  )
```

```r
survey_config <- list(
  lapop = list(
    trust_items = c(
      "justice_system", "electoral_tribunal", "armed_forces",
      "legislature", "public_ministry", "police", "auditor",
      "political_parties", "supreme_court", "municipality",
      "media", "elections"),
    cfa_syntax = '
      Trust =~ justice_system + electoral_tribunal +
               armed_forces + legislature + public_ministry +
               police + auditor + political_parties +
               supreme_court + municipality + media + elections
      armed_forces ~~ legislature
      public_ministry ~~ auditor
    ',
    trust_ascending = TRUE),
  lb = list(
    trust_items = c(
      "Congress", "Natl_Govt", "Judiciary", "Parties",
      "Electoral_Inst", "President"),
    cfa_syntax = '
      Political =~ Congress + Natl_Govt + Judiciary + Parties +
                   Electoral_Inst + President
      Natl_Govt ~~ President
    ',
    trust_ascending = FALSE)
)

cat("Combined (no Gemma):", nrow(all_results), "rows\n")
```

```
Combined (no Gemma): 75014 rows
```

Helper Functions

```r
#   CFA: fit and score
fit_cfa_score <- function(data_mat, cfa_syntax, row_ids) {
  tryCatch({
    fit <- cfa(cfa_syntax, data = data_mat,
               estimator = "ML", meanstructure = TRUE)
    fs <- as.numeric(lavPredict(fit, method = "regression"))
    omega <- as.numeric(semTools::compRelSEM(fit))
    list(scores = tibble(row_id = row_ids, fs = fs),
         omega = omega)
  }, error = function(e) {
```

```r
    cat("  CFA failed:", conditionMessage(e), "\n")
    list(scores = tibble(row_id = row_ids,
                         fs = rep(NA_real_, length(row_ids))),
         omega = NA_real_)
  })
}

#   IRT: fit and score (reverse-codes LB)
fit_irt_score <- function(data_mat, trust_ascending, row_ids) {
  item_max <- max(data_mat, na.rm = TRUE)
  irt_data <- if (!trust_ascending) {
    data_mat |>
      mutate(across(everything(),
                    function(x) as.integer(item_max + 1L - x)))
  } else {
    data_mat |>
      mutate(across(everything(), as.integer))
  }
  tryCatch({
    fit <- mirt(irt_data, 1, itemtype = "graded",
                verbose = FALSE)
    theta <- as.numeric(fscores(fit, method = "EAP")[, 1])
    list(scores = tibble(row_id = row_ids, theta = theta))
  }, error = function(e) {
    cat("  IRT failed:", conditionMessage(e), "\n")
    list(scores = tibble(row_id = row_ids,
                         theta = rep(NA_real_,
                                     length(row_ids))))
  })
}

#   Combined fit: CFA + IRT + correlation matrix
fit_both_models <- function(data_df, cfg, label = "") {
  trust_items <- cfg$trust_items
  row_ids <- data_df$row_id
  data_mat <- data_df |>
    select(all_of(trust_items)) |>
    mutate(across(everything(), as.numeric))
  cat("  Fitting:", label, "| N =", nrow(data_mat), "\n")
  cfa_out <- fit_cfa_score(data_mat, cfg$cfa_syntax, row_ids)
  irt_out <- fit_irt_score(data_mat, cfg$trust_ascending,
                           row_ids)
  cor_mat <- cor(data_mat, use = "pairwise.complete.obs")
  scores <- cfa_out$scores |>
    left_join(irt_out$scores, by = "row_id")
  list(scores = scores, omega = cfa_out$omega,
       cor_mat = cor_mat,
       mean_theta = mean(scores$theta, na.rm = TRUE),
```

```r
      sd_theta = sd(scores$theta, na.rm = TRUE),
      n = nrow(scores))
}

#   Reconstruct training + imputed test
reconstruct_full_data <- function(survey_name, fold_i,
                                   method_name, tier_name) {
  cfg <- survey_config[[survey_name]]
  train_df <- folds_list[[survey_name]] |>
    filter(fold_id == fold_i, split == "train") |>
    select(row_id, all_of(cfg$trust_items)) |>
    mutate(across(all_of(cfg$trust_items), as.numeric))
  test_df <- folds_list[[survey_name]] |>
    filter(fold_id == fold_i, split == "test") |>
    select(row_id, all_of(cfg$trust_items)) |>
    mutate(across(all_of(cfg$trust_items), as.numeric))
  preds <- all_results |>
    filter(survey == survey_name, fold_id == fold_i,
           method == method_name, tier == tier_name) |>
    select(row_id, item, predicted) |>
    mutate(predicted = as.numeric(predicted))
  if (nrow(preds) == 0L) return(NULL)
  preds_wide <- preds |>
    pivot_wider(names_from = item, values_from = predicted,
                names_prefix = "imp_")
  test_merged <- test_df |>
    left_join(preds_wide, by = "row_id")
  test_imputed <- map_dfc(cfg$trust_items, function(it) {
    actual_val <- test_merged[[it]]
    imp_col <- paste0("imp_", it)
    imp_val <- if (imp_col %in% names(test_merged)) {
      test_merged[[imp_col]]
    } else { NA_real_ }
    tibble(!!it := if_else(!is.na(imp_val),
                           imp_val, actual_val))
  }) |>
    mutate(row_id = test_merged$row_id)
  bind_rows(train_df, test_imputed)
}

#   Weighted kappa (quadratic weights)
compute_weighted_kappa <- function(actual, predicted) {
  cats <- sort(unique(c(actual, predicted)))
  k <- length(cats)
  if (k < 2) return(NA_real_)
  obs_tbl <- table(factor(actual, levels = cats),
                   factor(predicted, levels = cats))
  n <- sum(obs_tbl)
```

```r
  row_m <- rowSums(obs_tbl)
  col_m <- colSums(obs_tbl)
  exp_tbl <- outer(row_m, col_m) / n
  w <- outer(seq_along(cats), seq_along(cats),
             function(i, j) (i - j)^2 / (k - 1)^2)
  1 - sum(w * obs_tbl) / sum(w * exp_tbl)
}

#   IRT: fit and extract item parameters
# Returns discrimination (a) and raw intercepts (d) per item
# Uses coef(fit, printSE = TRUE) pattern from Script 02
fit_irt_extract <- function(data_df, cfg) {
  items <- cfg$trust_items
  data_mat <- data_df |>
    select(all_of(items)) |>
    mutate(across(everything(), as.numeric))
  item_max <- max(data_mat, na.rm = TRUE)
  irt_data <- if (!cfg$trust_ascending) {
    data_mat |>
      mutate(across(everything(),
                    function(x) as.integer(item_max + 1L - x)))
  } else {
    data_mat |>
      mutate(across(everything(), as.integer))
  }

  fit <- mirt(irt_data, 1, itemtype = "graded",
              verbose = FALSE)
  item_coefs <- coef(fit, printSE = TRUE)

  map_dfr(items, function(item_name) {
    mat <- item_coefs[[item_name]]
    par_row <- mat["par", ]
    param_names <- names(par_row)

    # Discrimination
    a_val <- par_row[["a1"]]

    # Raw intercepts (d) - NOT converted to b
    d_cols <- param_names[grepl("^d", param_names)]

    # Also compute b = -d/a for reference
    bind_rows(
      tibble(item = item_name, param = "a", value = a_val),
      map_dfr(d_cols, function(dc) {
        d_val <- par_row[[dc]]
        b_label <- gsub("^d", "b", dc)
        tibble(item = item_name, param = dc, value = d_val)
```

```
      })
    )
  })
}
```

Model Fitting

```
tic("Gold standard fits")
gold_fits <- map_dfr(names(survey_config), function(s) {
  cfg <- survey_config[[s]]
  map_dfr(1:5, function(fi) {
    full_df <- folds_list[[s]] |>
      filter(fold_id == fi) |>
      select(row_id, all_of(cfg$trust_items)) |>
      mutate(across(all_of(cfg$trust_items), as.numeric))
    result <- fit_both_models(
      full_df, cfg, paste("Gold", s, "fold", fi))
    tibble(survey = s, fold_id = fi,
           condition = "Gold",
           method_label = "Gold Standard",
           omega = result$omega,
           mean_theta = result$mean_theta,
           sd_theta = result$sd_theta,
           n = result$n,
           scores = list(result$scores),
           cor_mat = list(result$cor_mat))
  })
})
```

```
  Fitting: Gold lapop fold 1 | N = 1430
  Fitting: Gold lapop fold 2 | N = 1430
  Fitting: Gold lapop fold 3 | N = 1430
  Fitting: Gold lapop fold 4 | N = 1430
  Fitting: Gold lapop fold 5 | N = 1430
  Fitting: Gold lb fold 1 | N = 1073
  Fitting: Gold lb fold 2 | N = 1073
  Fitting: Gold lb fold 3 | N = 1073
  Fitting: Gold lb fold 4 | N = 1073
  Fitting: Gold lb fold 5 | N = 1073
```

```
toc()
```

```
Gold standard fits: 9.14 sec elapsed
```

```r
tic("Deletion fits")
deletion_fits <- map_dfr(names(survey_config), function(s) {
  cfg <- survey_config[[s]]
  map_dfr(1:5, function(fi) {
    train_df <- folds_list[[s]] |>
      filter(fold_id == fi, split == "train") |>
      select(row_id, all_of(cfg$trust_items)) |>
      mutate(across(all_of(cfg$trust_items), as.numeric))
    result <- fit_both_models(
      train_df, cfg, paste("Deletion", s, "fold", fi))
    tibble(survey = s, fold_id = fi,
           condition = "Deletion",
           method_label = "Listwise Deletion",
           omega = result$omega,
           mean_theta = result$mean_theta,
           sd_theta = result$sd_theta,
           n = result$n,
           scores = list(result$scores),
           cor_mat = list(result$cor_mat))
  })
})
```

```
  Fitting: Deletion lapop fold 1 | N = 1144
  Fitting: Deletion lapop fold 2 | N = 1144
  Fitting: Deletion lapop fold 3 | N = 1144
  Fitting: Deletion lapop fold 4 | N = 1144
  Fitting: Deletion lapop fold 5 | N = 1144
  Fitting: Deletion lb fold 1 | N = 858
  Fitting: Deletion lb fold 2 | N = 858
  Fitting: Deletion lb fold 3 | N = 858
  Fitting: Deletion lb fold 4 | N = 859
  Fitting: Deletion lb fold 5 | N = 859
```

```r
toc()
```

```
Deletion fits: 7.39 sec elapsed
```

```r
tic("All imputed fits")
method_tier_grid <- all_results |>
  distinct(survey, method, method_label, method_type, tier)
cat("Fitting", nrow(method_tier_grid),
    "method-tier combos × 5 folds\n")
```

```
Fitting 24 method-tier combos × 5 folds
```

```r
imputed_fits <- pmap_dfr(method_tier_grid, function(
    survey, method, method_label, method_type, tier) {
  cfg <- survey_config[[survey]]
  map_dfr(1:5, function(fi) {
    recon_df <- reconstruct_full_data(
      survey, fi, method, as.character(tier))
    if (is.null(recon_df)) {
      return(tibble(
        survey = survey, fold_id = fi,
        condition = "Imputed",
        method = method, method_label = method_label,
        method_type = as.character(method_type),
        tier = as.character(tier),
        omega = NA_real_,
        mean_theta = NA_real_, sd_theta = NA_real_,
        n = 0L, scores = list(NULL),
        cor_mat = list(NULL)))
    }
    result <- fit_both_models(
      recon_df, cfg,
      paste(method_label, tier, survey, "fold", fi))
    tibble(survey = survey, fold_id = fi,
           condition = "Imputed",
           method = method, method_label = method_label,
           method_type = as.character(method_type),
           tier = as.character(tier),
           omega = result$omega,
           mean_theta = result$mean_theta,
           sd_theta = result$sd_theta,
           n = result$n,
           scores = list(result$scores),
           cor_mat = list(result$cor_mat))
  })
})
```

```
Fitting: mixgb S lapop fold 1 | N = 1430
Fitting: mixgb S lapop fold 2 | N = 1430
Fitting: mixgb S lapop fold 3 | N = 1430
Fitting: mixgb S lapop fold 4 | N = 1430
Fitting: mixgb S lapop fold 5 | N = 1430
Fitting: miceRanger S lapop fold 1 | N = 1430
Fitting: miceRanger S lapop fold 2 | N = 1430
Fitting: miceRanger S lapop fold 3 | N = 1430
Fitting: miceRanger S lapop fold 4 | N = 1430
Fitting: miceRanger S lapop fold 5 | N = 1430
Fitting: mixgb P1 lapop fold 1 | N = 1430
Fitting: mixgb P1 lapop fold 2 | N = 1430
```

```
Fitting: mixgb P1 lapop fold 3 | N = 1430
Fitting: mixgb P1 lapop fold 4 | N = 1430
Fitting: mixgb P1 lapop fold 5 | N = 1430
Fitting: miceRanger P1 lapop fold 1 | N = 1430
Fitting: miceRanger P1 lapop fold 2 | N = 1430
Fitting: miceRanger P1 lapop fold 3 | N = 1430
Fitting: miceRanger P1 lapop fold 4 | N = 1430
Fitting: miceRanger P1 lapop fold 5 | N = 1430
Fitting: mixgb P2 lapop fold 1 | N = 1430
Fitting: mixgb P2 lapop fold 2 | N = 1430
Fitting: mixgb P2 lapop fold 3 | N = 1430
Fitting: mixgb P2 lapop fold 4 | N = 1430
Fitting: mixgb P2 lapop fold 5 | N = 1430
Fitting: miceRanger P2 lapop fold 1 | N = 1430
Fitting: miceRanger P2 lapop fold 2 | N = 1430
Fitting: miceRanger P2 lapop fold 3 | N = 1430
Fitting: miceRanger P2 lapop fold 4 | N = 1430
Fitting: miceRanger P2 lapop fold 5 | N = 1430
Fitting: mixgb S lb fold 1 | N = 1073
Fitting: mixgb S lb fold 2 | N = 1073
Fitting: mixgb S lb fold 3 | N = 1073
Fitting: mixgb S lb fold 4 | N = 1073
Fitting: mixgb S lb fold 5 | N = 1073
Fitting: miceRanger S lb fold 1 | N = 1073
Fitting: miceRanger S lb fold 2 | N = 1073
Fitting: miceRanger S lb fold 3 | N = 1073
Fitting: miceRanger S lb fold 4 | N = 1073
Fitting: miceRanger S lb fold 5 | N = 1073
Fitting: mixgb P1 lb fold 1 | N = 1073
Fitting: mixgb P1 lb fold 2 | N = 1073
Fitting: mixgb P1 lb fold 3 | N = 1073
Fitting: mixgb P1 lb fold 4 | N = 1073
Fitting: mixgb P1 lb fold 5 | N = 1073
Fitting: miceRanger P1 lb fold 1 | N = 1073
Fitting: miceRanger P1 lb fold 2 | N = 1073
Fitting: miceRanger P1 lb fold 3 | N = 1073
Fitting: miceRanger P1 lb fold 4 | N = 1073
Fitting: miceRanger P1 lb fold 5 | N = 1073
Fitting: mixgb P2 lb fold 1 | N = 1073
Fitting: mixgb P2 lb fold 2 | N = 1073
Fitting: mixgb P2 lb fold 3 | N = 1073
Fitting: mixgb P2 lb fold 4 | N = 1073
Fitting: mixgb P2 lb fold 5 | N = 1073
Fitting: miceRanger P2 lb fold 1 | N = 1073
Fitting: miceRanger P2 lb fold 2 | N = 1073
Fitting: miceRanger P2 lb fold 3 | N = 1073
Fitting: miceRanger P2 lb fold 4 | N = 1073
Fitting: miceRanger P2 lb fold 5 | N = 1073
```

```
Fitting: Claude Sonnet P1 lapop fold 1 | N = 1430
Fitting: Claude Sonnet P1 lapop fold 2 | N = 1430
Fitting: Claude Sonnet P1 lapop fold 3 | N = 1430
Fitting: Claude Sonnet P1 lapop fold 4 | N = 1430
Fitting: Claude Sonnet P1 lapop fold 5 | N = 1430
Fitting: Claude Sonnet P2 lapop fold 1 | N = 1430
Fitting: Claude Sonnet P2 lapop fold 2 | N = 1430
Fitting: Claude Sonnet P2 lapop fold 3 | N = 1430
Fitting: Claude Sonnet P2 lapop fold 4 | N = 1430
Fitting: Claude Sonnet P2 lapop fold 5 | N = 1430
Fitting: Claude Sonnet S lapop fold 1 | N = 1430
Fitting: Claude Sonnet S lapop fold 2 | N = 1430
Fitting: Claude Sonnet S lapop fold 3 | N = 1430
Fitting: Claude Sonnet S lapop fold 4 | N = 1430
Fitting: Claude Sonnet S lapop fold 5 | N = 1430
Fitting: Maverick P1 lapop fold 1 | N = 1430
Fitting: Maverick P1 lapop fold 2 | N = 1430
Fitting: Maverick P1 lapop fold 3 | N = 1430
Fitting: Maverick P1 lapop fold 4 | N = 1430
Fitting: Maverick P1 lapop fold 5 | N = 1430
Fitting: Maverick P2 lapop fold 1 | N = 1430
Fitting: Maverick P2 lapop fold 2 | N = 1430
Fitting: Maverick P2 lapop fold 3 | N = 1430
Fitting: Maverick P2 lapop fold 4 | N = 1430
Fitting: Maverick P2 lapop fold 5 | N = 1430
Fitting: Maverick S lapop fold 1 | N = 1430
Fitting: Maverick S lapop fold 2 | N = 1430
Fitting: Maverick S lapop fold 3 | N = 1430
Fitting: Maverick S lapop fold 4 | N = 1430
Fitting: Maverick S lapop fold 5 | N = 1430
Fitting: Claude Sonnet P1 lb fold 1 | N = 1073
Fitting: Claude Sonnet P1 lb fold 2 | N = 1073
Fitting: Claude Sonnet P1 lb fold 3 | N = 1073
Fitting: Claude Sonnet P1 lb fold 4 | N = 1073
Fitting: Claude Sonnet P1 lb fold 5 | N = 1073
Fitting: Claude Sonnet P2 lb fold 1 | N = 1073
Fitting: Claude Sonnet P2 lb fold 2 | N = 1073
Fitting: Claude Sonnet P2 lb fold 3 | N = 1073
Fitting: Claude Sonnet P2 lb fold 4 | N = 1073
Fitting: Claude Sonnet P2 lb fold 5 | N = 1073
Fitting: Claude Sonnet S lb fold 1 | N = 1073
Fitting: Claude Sonnet S lb fold 2 | N = 1073
Fitting: Claude Sonnet S lb fold 3 | N = 1073
Fitting: Claude Sonnet S lb fold 4 | N = 1073
Fitting: Claude Sonnet S lb fold 5 | N = 1073
Fitting: Maverick P1 lb fold 1 | N = 1073
Fitting: Maverick P1 lb fold 2 | N = 1073
Fitting: Maverick P1 lb fold 3 | N = 1073
```

```
  Fitting: Maverick P1 lb fold 4 | N = 1073
  Fitting: Maverick P1 lb fold 5 | N = 1073
  Fitting: Maverick P2 lb fold 1 | N = 1073
  Fitting: Maverick P2 lb fold 2 | N = 1073
  Fitting: Maverick P2 lb fold 3 | N = 1073
  Fitting: Maverick P2 lb fold 4 | N = 1073
  Fitting: Maverick P2 lb fold 5 | N = 1073
  Fitting: Maverick S lb fold 1 | N = 1073
  Fitting: Maverick S lb fold 2 | N = 1073
  Fitting: Maverick S lb fold 3 | N = 1073
  Fitting: Maverick S lb fold 4 | N = 1073
  Fitting: Maverick S lb fold 5 | N = 1073
```

```r
toc()
```

```
All imputed fits: 106.89 sec elapsed
```

RQ1: Psychometric Context Utilization

Item-Level Metrics

```r
rq1_item <- all_results |>
  group_by(survey, survey_label, method, method_label,
           method_type, tier, fold_id) |>
  summarise(
    accuracy = mean(predicted == actual),
    mae = mean(abs(predicted - actual)),
    wkappa = compute_weighted_kappa(actual, predicted),
    .groups = "drop")

rq1_item_summary <- rq1_item |>
  group_by(survey, survey_label, method, method_label,
           method_type, tier) |>
  summarise(
    accuracy = round(mean(accuracy), 3),
    mae = round(mean(mae), 3),
    wkappa = round(mean(wkappa), 3),
    .groups = "drop")
```

```r
rq1_item_summary |>
  select(Survey = survey_label, Type = method_type,
         Method = method_label, Tier = tier,
         Acc = accuracy, MAE = mae, `W ` = wkappa) |>
  arrange(Survey, Type, Method, Tier) |>
  knitr::kable(align = "lllcccc")
```

Table 1: RQ1: Item-Level Metrics — All Methods × Tiers (5-Fold CV). W = quadratic weighted kappa.

Survey	Type	Method	Tier	Acc	MAE	W
LAPOP Mexico	Traditional	miceRanger	S	0.329	1.205	0.561
LAPOP Mexico	Traditional	miceRanger	P1	0.343	1.141	0.592
LAPOP Mexico	Traditional	miceRanger	P2	0.344	1.144	0.595
LAPOP Mexico	Traditional	mixgb	S	0.382	1.058	0.615
LAPOP Mexico	Traditional	mixgb	P1	0.393	1.004	0.647
LAPOP Mexico	Traditional	mixgb	P2	0.393	0.989	0.661
LAPOP Mexico	LLM	Claude Sonnet	S	0.359	0.968	0.645
LAPOP Mexico	LLM	Claude Sonnet	P1	0.379	0.956	0.667
LAPOP Mexico	LLM	Claude Sonnet	P2	0.394	0.960	0.675
LAPOP Mexico	LLM	Maverick	S	0.366	1.025	0.634
LAPOP Mexico	LLM	Maverick	P1	0.371	1.042	0.639
LAPOP Mexico	LLM	Maverick	P2	0.367	1.055	0.640

Table 1: RQ1: Item-Level Metrics — All Methods × Tiers (5-Fold CV). W = quadratic weighted kappa.

Survey	Type	Method	Tier	Acc	MAE	W
Latinobarómetro Colombia	Traditional	miceRanger	S	0.552	0.534	0.504
Latinobarómetro Colombia	Traditional	miceRanger	P1	0.558	0.524	0.516
Latinobarómetro Colombia	Traditional	miceRanger	P2	0.574	0.506	0.526
Latinobarómetro Colombia	Traditional	mixgb	S	0.566	0.518	0.463
Latinobarómetro Colombia	Traditional	mixgb	P1	0.574	0.499	0.520
Latinobarómetro Colombia	Traditional	mixgb	P2	0.589	0.483	0.541
Latinobarómetro Colombia	LLM	Claude Sonnet	S	0.598	0.460	0.550
Latinobarómetro Colombia	LLM	Claude Sonnet	P1	0.603	0.455	0.559
Latinobarómetro Colombia	LLM	Claude Sonnet	P2	0.614	0.444	0.579
Latinobarómetro Colombia	LLM	Maverick	S	0.595	0.468	0.528
Latinobarómetro Colombia	LLM	Maverick	P1	0.581	0.513	0.530
Latinobarómetro Colombia	LLM	Maverick	P2	0.592	0.485	0.548

Figure: Tier Effect on Item Recovery

```r
ggplot(rq1_item_summary,
       aes(x = tier, y = wkappa,
           color = method_label,
           group = method_label,
           linetype = method_type)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2.5) +
  facet_wrap(~ survey_label, scales = "free_y") +
  scale_color_viridis_d(option = "D", end = 0.85) +
  labs(x = "Information Tier",
       y = "Weighted Kappa",
       color = "Method", linetype = "Type") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom",
```

```
    strip.text = element_text(face = "bold"))
```

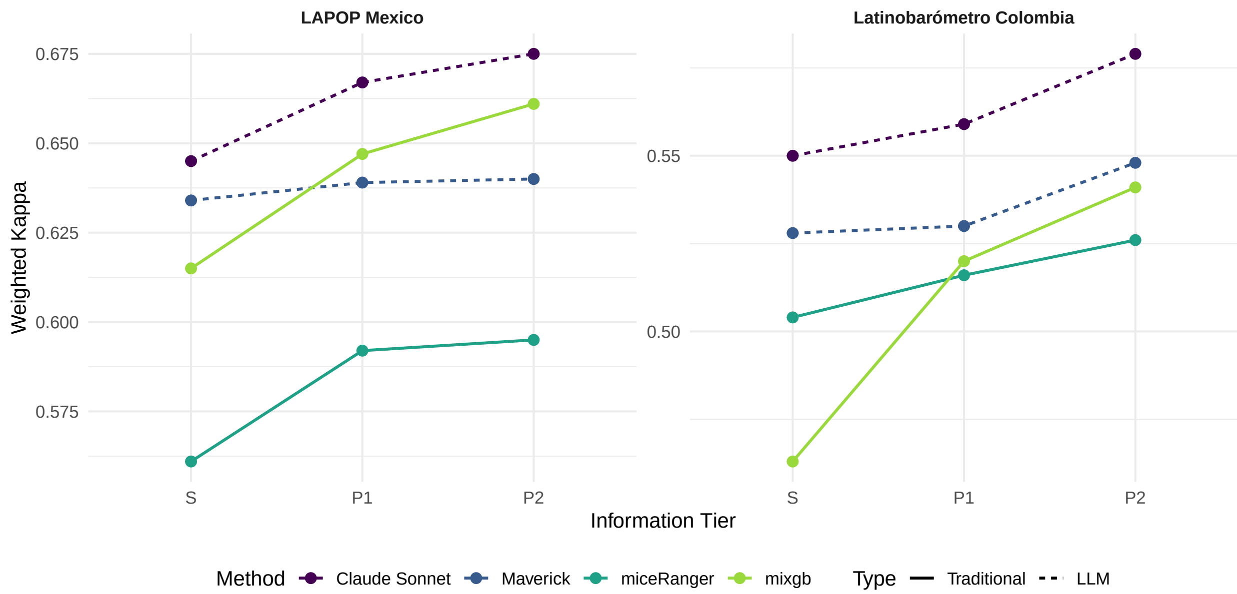

Figure 1: RQ1: Item-Level Performance by Information Tier. All methods, both surveys. Upward trajectory from S to P2 confirms psychometric context improves item recovery.

RQ2: Construct-Level Recovery

Person-Level Matching

```
# Match imputed to gold for test respondents
rq2_person <- pmap_dfr(
  imputed_fits |>
    filter(!map_lgl(scores, is.null)) |>
    select(survey, fold_id, method, method_label,
           method_type, tier, scores),
  function(survey, fold_id, method, method_label,
           method_type, tier, scores) {
    gold_scores <- gold_fits |>
      filter(survey == !!survey,
             fold_id == !!fold_id) |>
      pull(scores) |> pluck(1)
    if (is.null(gold_scores)) return(NULL)
    test_ids <- folds_list[[survey]] |>
      filter(fold_id == !!fold_id, split == "test") |>
      pull(row_id)
    gold_scores |>
      filter(row_id %in% test_ids) |>
      rename(theta_gold = theta, fs_gold = fs) |>
      inner_join(
        scores |> rename(theta_imp = theta, fs_imp = fs),
        by = "row_id") |>
      mutate(survey = survey, fold_id = fold_id,
             method = method, method_label = method_label,
             method_type = method_type, tier = tier)
  }
) |>
  mutate(survey_label = if_else(survey == "lapop",
    "LAPOP Mexico", "Latinobarómetro Colombia"))

cat("RQ2 person-level:", nrow(rq2_person), "rows\n")
```

```
RQ2 person-level: 30036 rows
```

Correlation and Slope

```
rq2_metrics <- rq2_person |>
  group_by(survey, survey_label, method, method_label,
           method_type, tier, fold_id) |>
  summarise(
```

```r
    r = cor(theta_gold, theta_imp, use = "complete.obs"),
    slope = coef(lm(theta_imp ~ theta_gold))[["theta_gold"]],
    .groups = "drop")

rq2_summary <- rq2_metrics |>
  group_by(survey_label, method_type, method_label, tier) |>
  summarise(
    r = round(mean(r, na.rm = TRUE), 4),
    slope = round(mean(slope, na.rm = TRUE), 4),
    .groups = "drop")
```

```r
rq2_summary |>
  select(Survey = survey_label, Type = method_type,
         Method = method_label, Tier = tier,
         r, Slope = slope) |>
  arrange(Survey, Tier, desc(r)) |>
  knitr::kable(align = "lllccc")
```

Table 2: RQ2: Construct Recovery — Correlation (r) and Regression Slope of _imputed on _gold. r = rank preservation; slope = scale fidelity (1.0 = perfect, <1 = compression).

Survey	Type	Method	Tier	r	Slope
LAPOP Mexico	LLM	Claude Sonnet	P1	0.9883	1.0187
LAPOP Mexico	Traditional	miceRanger	P1	0.9869	1.0100
LAPOP Mexico	Traditional	mixgb	P1	0.9869	1.0222
LAPOP Mexico	LLM	Maverick	P1	0.9845	1.0382
LAPOP Mexico	LLM	Claude Sonnet	P2	0.9883	1.0336
LAPOP Mexico	Traditional	mixgb	P2	0.9879	1.0240
LAPOP Mexico	Traditional	miceRanger	P2	0.9870	1.0138
LAPOP Mexico	LLM	Maverick	P2	0.9837	1.0354
LAPOP Mexico	LLM	Claude Sonnet	S	0.9891	0.9949
LAPOP Mexico	LLM	Maverick	S	0.9860	1.0259
LAPOP Mexico	Traditional	mixgb	S	0.9850	1.0121
LAPOP Mexico	Traditional	miceRanger	S	0.9836	0.9986
Latinobarómetro Colombia	Traditional	mixgb	P1	0.9204	0.9417
Latinobarómetro Colombia	LLM	Claude Sonnet	P1	0.9202	1.0007
Latinobarómetro Colombia	Traditional	miceRanger	P1	0.9177	0.9428
Latinobarómetro Colombia	LLM	Maverick	P1	0.9054	1.0311
Latinobarómetro Colombia	Traditional	mixgb	P2	0.9292	0.9899
Latinobarómetro Colombia	LLM	Claude Sonnet	P2	0.9245	1.0096
Latinobarómetro Colombia	Traditional	miceRanger	P2	0.9218	0.9784
Latinobarómetro Colombia	LLM	Maverick	P2	0.9140	1.0147
Latinobarómetro Colombia	LLM	Claude Sonnet	S	0.9300	0.9898
Latinobarómetro Colombia	LLM	Maverick	S	0.9192	0.9934
Latinobarómetro Colombia	Traditional	miceRanger	S	0.9076	0.9343

Table 2: RQ2: Construct Recovery — Correlation (r) and Regression Slope of _imputed on _gold. r = rank preservation; slope = scale fidelity (1.0 = perfect, <1 = compression).

Survey	Type	Method	Tier	r	Slope
Latinobarómetro Colombia	Traditional	mixgb	S	0.8951	0.8887

Figure: Recovery Scatterplots

```
scatter_data <- rq2_person |>
  filter(fold_id == 1,
         (method == "claude_sonnet" & tier == "P2") |
         (method == "mixgb" & tier == "S")) |>
  mutate(label = paste(method_label,
                       if_else(method_type == "LLM",
                               "(P2)", "(S)")))

ggplot(scatter_data,
       aes(x = theta_gold, y = theta_imp)) +
  geom_abline(slope = 1, intercept = 0,
              linetype = "dashed", color = "grey50",
              linewidth = 0.5) +
  geom_point(alpha = 0.15, size = 0.8,
             color = viridis(3)[2]) +
  geom_smooth(method = "lm", se = FALSE,
              color = viridis(3)[1],
              linewidth = 0.9) +
  facet_grid(label ~ survey_label, scales = "free") +
  labs(x = expression(theta[gold]),
       y = expression(theta[imputed])) +
  theme_minimal(base_size = 11) +
  theme(strip.text = element_text(face = "bold"))
```

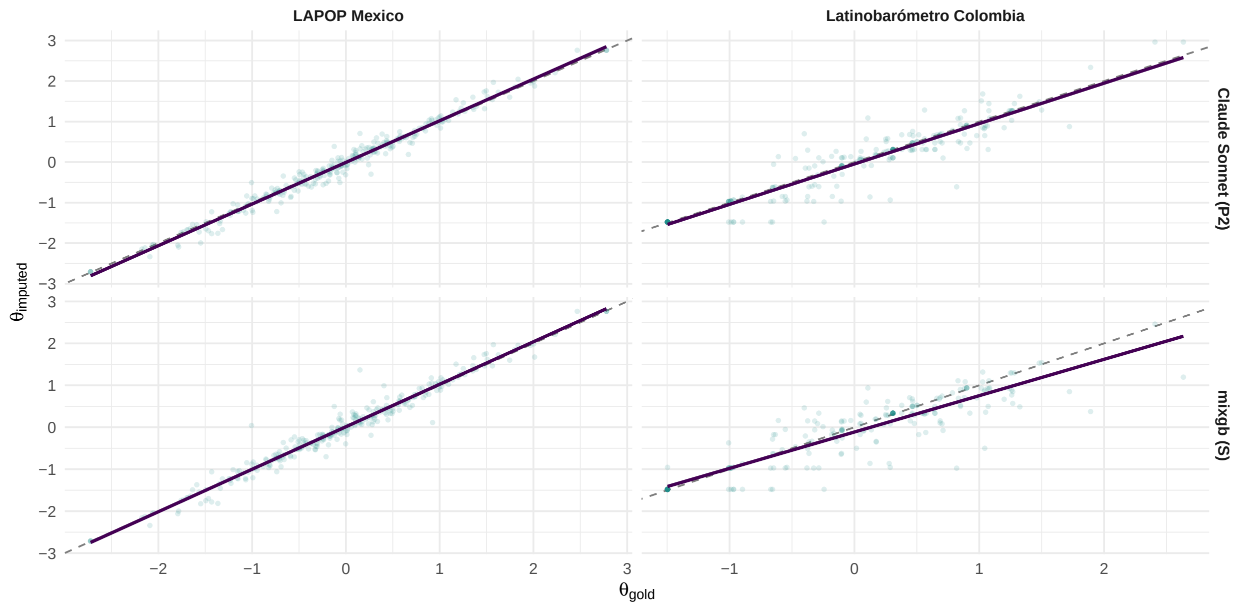

Figure 2: RQ2: Construct Recovery — _gold vs. _imputed for Claude P2 and mixgb S (Fold 1). Solid line = regression fit; dashed = identity. Deviation from slope = 1 indicates compression or expansion.

Figure: Correlation by Method and Tier

```r
ggplot(rq2_summary,
       aes(x = tier, y = r,
           color = method_label,
           group = method_label,
           linetype = method_type)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2.5) +
  facet_wrap(~ survey_label, scales = "free_y") +
  scale_color_viridis_d(option = "D", end = 0.85) +
  labs(x = "Information Tier",
       y = "Correlation (r) with Gold Standard  ",
       color = "Method", linetype = "Type") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom",
        strip.text = element_text(face = "bold"))
```

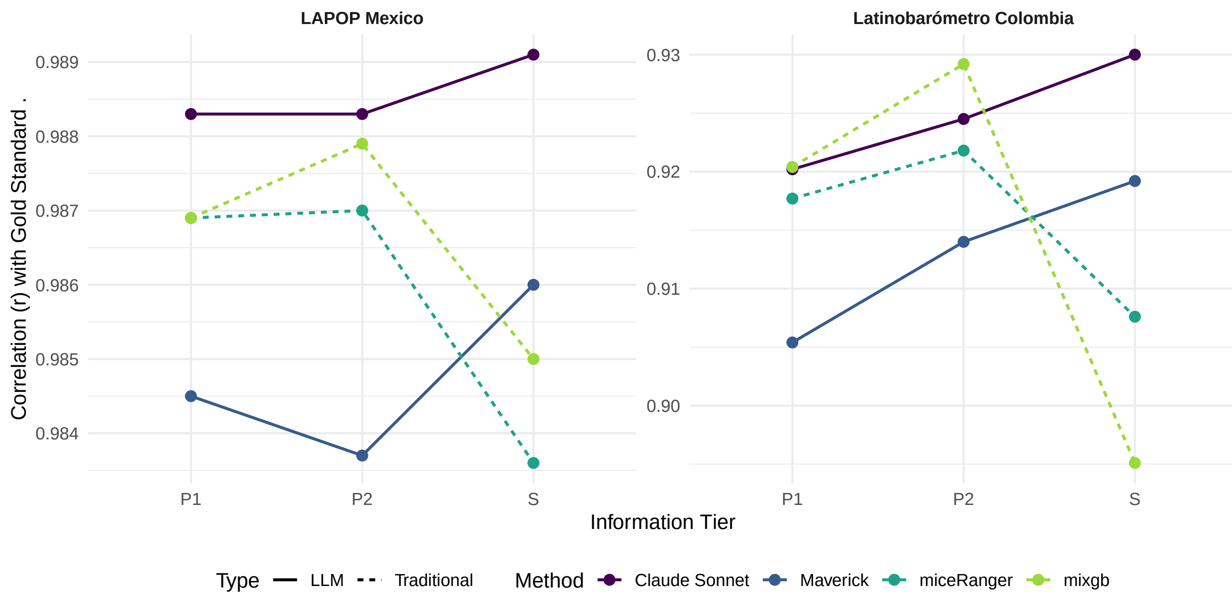

Figure 3: RQ2: Rank Preservation (r) by Method and Tier. All methods on both surveys.

RQ3: Item Measurement Error

RQ3 asks: what does imputation do to the measurement model's item parameters? We refit IRT-GRM on four conditions per fold (gold, deletion, Claude P2, mixgb S), extract discrimination (a) and raw intercept (d) parameters, and decompose per-item MSE into $\text{Bias}^2 + \text{Variance}$ following Biemer (2010). Item-level response variance is also compared across conditions.

Extract IRT Item Parameters

```r
tic("IRT parameter extraction")
rq3_irt_params <- map_dfr(c("lapop", "lb"), function(s) {
  cfg <- survey_config[[s]]
  map_dfr(1:5, function(fi) {
    cat("  Extracting:", s, "fold", fi, "\n")

    # Gold: full fold (training + test, no masking)
    gold_df <- folds_list[[s]] |>
      filter(fold_id == fi) |>
      select(row_id, all_of(cfg$trust_items)) |>
      mutate(across(all_of(cfg$trust_items), as.numeric))
    gold_p <- fit_irt_extract(gold_df, cfg) |>
      mutate(condition = "Gold")

    # Deletion: training only (listwise deletion of test fold)
    del_df <- folds_list[[s]] |>
      filter(fold_id == fi, split == "train") |>
      select(row_id, all_of(cfg$trust_items)) |>
      mutate(across(all_of(cfg$trust_items), as.numeric))
    del_p <- fit_irt_extract(del_df, cfg) |>
      mutate(condition = "Deletion")

    # Claude Sonnet P2 (best LLM configuration)
    claude_df <- reconstruct_full_data(
      s, fi, "claude_sonnet", "P2")
    claude_p <- fit_irt_extract(claude_df, cfg) |>
      mutate(condition = "Claude P2")

    # mixgb S (traditional benchmark - fair comparison tier)
    mixgb_df <- reconstruct_full_data(s, fi, "mixgb", "S")
    mixgb_p <- if (!is.null(mixgb_df)) {
      fit_irt_extract(mixgb_df, cfg) |>
        mutate(condition = "mixgb S")
    } else {
      tibble(item = character(), param = character(),
             value = numeric(), condition = character())
    }
```

```
    bind_rows(gold_p, del_p, claude_p, mixgb_p) |>
      mutate(survey = s, fold_id = fi)
  })
}) |>
  mutate(
    survey_label = if_else(survey == "lapop",
      "LAPOP Mexico", "Latinobarómetro Colombia"),
    condition = factor(condition,
      levels = c("Gold", "Deletion", "Claude P2", "mixgb S"))
  )
```

```
Extracting: lapop fold 1
Extracting: lapop fold 2
Extracting: lapop fold 3
Extracting: lapop fold 4
Extracting: lapop fold 5
Extracting: lb fold 1
Extracting: lb fold 2
Extracting: lb fold 3
Extracting: lb fold 4
Extracting: lb fold 5
```

```
toc()
```

```
IRT parameter extraction: 31.5 sec elapsed
```

```
cat("RQ3 IRT params:", nrow(rq3_irt_params), "rows\n")
```

```
RQ3 IRT params: 2160 rows
```

Discrimination (a): Biemer Decomposition

```
#   Per item × fold: Δa for all conditions vs Gold
rq3_delta_a <- rq3_irt_params |>
  filter(param == "a") |>
  select(survey_label, item, fold_id, condition, value) |>
  pivot_wider(names_from = condition,
              values_from = value) |>
  mutate(
    delta_claude = `Claude P2` - Gold,
    delta_del    = Deletion - Gold,
```

```r
    delta_mixgb   = `mixgb S` - Gold
  )

#   Helper: Biemer decomposition for any delta column
# Returns per-item Bias, Bias², Variance, MSE, and % Syst
biemer_decompose <- function(data, delta_col, config_label) {
  data |>
    group_by(survey_label, item) |>
    summarise(
      gold_a    = round(mean(Gold), 3),
      bias      = round(mean(.data[[delta_col]]), 4),
      bias_sq   = mean(.data[[delta_col]])^2,
      variance = var(.data[[delta_col]]),
      mse       = mean(.data[[delta_col]]^2),
      pct_syst = round(
        if_else(mean(.data[[delta_col]]^2) > 0,
                mean(.data[[delta_col]])^2 /
                  mean(.data[[delta_col]]^2) * 100,
                0), 1),
      .groups = "drop"
    ) |>
    mutate(config = config_label) |>
    arrange(survey_label, desc(mse))
}

#   Apply to all three conditions
rq3_biemer_claude <- biemer_decompose(
  rq3_delta_a, "delta_claude", "Claude P2")
rq3_biemer_del <- biemer_decompose(
  rq3_delta_a, "delta_del", "Deletion")
rq3_biemer_mixgb <- biemer_decompose(
  rq3_delta_a, "delta_mixgb", "mixgb S")
```

Table: Item-Level Biemer — Claude P2 vs mixgb S

```r
#   Side-by-side item-level table
rq3_item_compare <- rq3_biemer_claude |>
  select(survey_label, item, gold_a,
         bias_c = bias, mse_c = mse, pct_c = pct_syst) |>
  left_join(
    rq3_biemer_mixgb |>
      select(survey_label, item,
             bias_m = bias, mse_m = mse, pct_m = pct_syst),
    by = c("survey_label", "item")
  ) |>
```

```
  arrange(survey_label, desc(gold_a))

rq3_item_compare |>
  mutate(
    bias_c = sprintf("%+.4f", bias_c),
    mse_c  = sprintf("%.4f", mse_c),
    bias_m = sprintf("%+.4f", bias_m),
    mse_m  = sprintf("%.4f", mse_m)
  ) |>
  select(Survey = survey_label, Item = item,
         `Gold a` = gold_a,
         `Claude Bias` = bias_c, `Claude MSE` = mse_c,
         `Claude %Sys` = pct_c,
         `mixgb Bias` = bias_m, `mixgb MSE` = mse_m,
         `mixgb %Sys` = pct_m) |>
  knitr::kable(align = "llcccccccc")
```

Table 3: RQ3: IRT Discrimination (a) — Biemer Decomposition. Claude P2 and mixgb S vs Gold Standard. Bias = mean Δa across 5 folds. MSE = Bias² + Variance.

Survey	Item	Gold a	Claude Bias	Claude MSE	Claude %Sys	mixgb Bias	mixgb MSE	mixgb %Sys
LAPOP Mexico	justice_system	3.304	+0.1611	0.0273	95.0	+0.0689	0.0064	74.3
LAPOP Mexico	auditor	2.968	+0.1614	0.0268	97.2	+0.0758	0.0065	89.1
LAPOP Mexico	public_ministry	2.792	+0.0973	0.0110	85.7	+0.0537	0.0037	79.1
LAPOP Mexico	supreme_court	2.548	+0.0962	0.0112	82.8	+0.0578	0.0039	85.9
LAPOP Mexico	legislature	2.263	+0.0632	0.0049	82.1	+0.0180	0.0006	57.2
LAPOP Mexico	elections	2.209	+0.0949	0.0103	87.0	+0.0630	0.0049	80.9
LAPOP Mexico	political_parties	2.133	+0.0603	0.0044	83.7	+0.0316	0.0017	60.1
LAPOP Mexico	electoral_tribunal	1.994	+0.0766	0.0065	90.6	+0.0450	0.0023	89.0
LAPOP Mexico	municipality	1.984	+0.0812	0.0075	87.8	+0.0584	0.0047	72.3
LAPOP Mexico	police	1.918	+0.0760	0.0063	91.7	+0.0413	0.0021	83.1
LAPOP Mexico	media	1.367	+0.0615	0.0043	88.8	+0.0312	0.0016	59.9
LAPOP Mexico	armed_forces	1.333	+0.0767	0.0060	98.0	+0.0275	0.0010	73.5

Table 3: RQ3: IRT Discrimination (a) — Biemer Decomposition. Claude P2 and mixgb S vs Gold Standard. Bias = mean Δa across 5 folds. MSE = Bias² + Variance.

Survey	Item	Gold a	Claude Bias	Claude MSE	Claude %Sys	mixgb Bias	mixgb MSE	mixgb %Sys
Latinobarómetro Colombia	Natl_Govt	2.662	+0.1343	0.0223	80.7	-0.0071	0.0065	0.8
Latinobarómetro Colombia	Judiciary	2.135	+0.2061	0.0492	86.3	+0.0531	0.0080	35.2
Latinobarómetro Colombia	Parties	2.121	+0.2443	0.0691	86.4	+0.0846	0.0162	44.1
Latinobarómetro Colombia	Electoral_Inst	2.035	+0.2433	0.0660	89.7	+0.0696	0.0104	46.7
Latinobarómetro Colombia	President	2.005	+0.1242	0.0216	71.4	+0.0278	0.0083	9.3
Latinobarómetro Colombia	Congress	1.993	+0.2232	0.0505	98.7	+0.0495	0.0041	59.8

Table: Aggregate MSE Summary (Three Conditions)

```
bind_rows(
  rq3_biemer_claude, rq3_biemer_mixgb, rq3_biemer_del
) |>
  group_by(survey_label, config) |>
  summarise(
    mean_bias    = round(mean(bias), 4),
    mean_bias_sq = round(mean(bias_sq), 6),
    mean_var     = round(mean(variance), 6),
    mean_mse     = round(mean(mse), 6),
    mean_pct     = round(mean(pct_syst), 1),
    .groups = "drop"
  ) |>
  mutate(
    config = factor(config,
      levels = c("Claude P2", "mixgb S", "Deletion"))
  ) |>
  select(Survey = survey_label, Condition = config,
         Bias = mean_bias, `Bias²` = mean_bias_sq,
         Variance = mean_var, MSE = mean_mse,
         `% Syst` = mean_pct) |>
  arrange(Survey, Condition) |>
  knitr::kable(align = "llcccc")
```

Table 4: RQ3: Discrimination MSE Summary — Claude P2, mixgb S, and Deletion. Mean across items per survey.

Survey	Condition	Bias	Bias²	Variance	MSE	% Syst
LAPOP Mexico	Claude P2	0.0922	0.009610	0.001166	0.010543	89.2
LAPOP Mexico	mixgb S	0.0477	0.002574	0.000867	0.003268	75.4
LAPOP Mexico	Deletion	0.0007	0.000001	0.004238	0.003392	0.0
Latinobarómetro Colombia	Claude P2	0.1959	0.040773	0.007103	0.046455	85.5
Latinobarómetro Colombia	mixgb S	0.0462	0.003017	0.007388	0.008927	32.6
Latinobarómetro Colombia	Deletion	0.0034	0.000014	0.015224	0.012193	0.1

Figure: Discrimination Shift Dumbbell (Claude P2 vs mixgb S)

```
#   Build dumbbell data
dumbbell_df <- rq3_biemer_claude |>
  select(survey_label, item, gold_a,
         bias_claude = bias) |>
  left_join(
    rq3_biemer_mixgb |>
      select(survey_label, item, bias_mixgb = bias),
    by = c("survey_label", "item")
  )

#   Viridis colors for the two methods
method_colors <- c("Claude P2" = viridis(4)[3],
                   "mixgb S"   = viridis(4)[1])

ggplot(dumbbell_df, aes(y = reorder(item, gold_a))) +
  # Zero reference line (gold standard = no distortion)
  geom_vline(xintercept = 0, linetype = "dashed",
             color = "grey50") +
  # Connecting segment between methods per item
  geom_segment(
    aes(x = bias_claude, xend = bias_mixgb,
        yend = reorder(item, gold_a)),
    color = "grey60", linewidth = 0.6
  ) +
  # Claude P2 points
  geom_point(aes(x = bias_claude, color = "Claude P2"),
             size = 3) +
  # mixgb S points
```

```r
geom_point(aes(x = bias_mixgb, color = "mixgb S"),
           size = 3) +
# Labels: Claude P2
geom_text(
  aes(x = bias_claude,
      label = sprintf("%+.3f", bias_claude)),
  hjust = -0.3, size = 2.5, color = viridis(4)[3]
) +
# Labels: mixgb S
geom_text(
  aes(x = bias_mixgb,
      label = sprintf("%+.3f", bias_mixgb)),
  hjust = 1.3, size = 2.5, color = viridis(4)[1]
) +
facet_wrap(~ survey_label, scales = "free") +
scale_color_manual(values = method_colors, name = NULL) +
labs(
  x = expression(Delta * "a (Condition - Gold)"),
  y = NULL,
  title = "Discrimination shift: Claude P2 vs mixgb S"
) +
theme_minimal(base_size = 12) +
theme(
  strip.text = element_text(face = "bold"),
  panel.grid.minor = element_blank(),
  legend.position = "bottom"
)
```

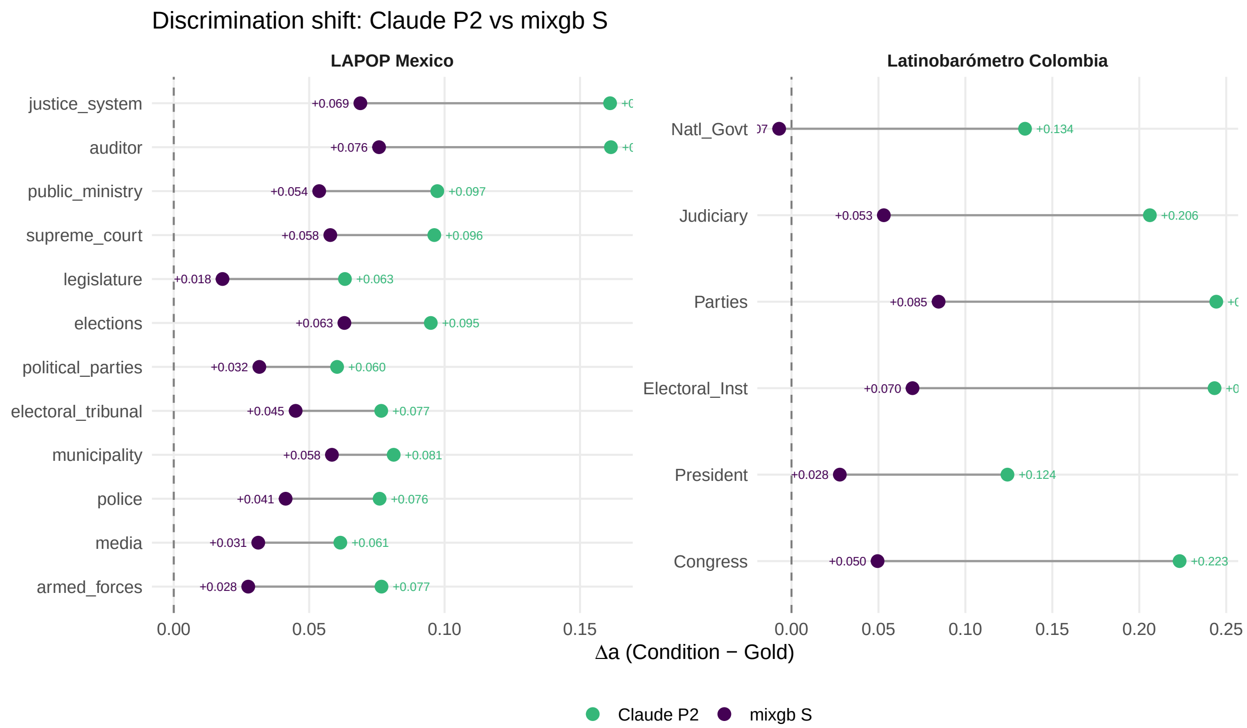

Figure 4: RQ3: IRT Discrimination Shift by Item — Claude P2 vs mixgb S. Each segment connects the two methods' mean Δa (condition − Gold) for the same item. Vertical dashed line = gold standard (no distortion).

Figure: Biemer Decomposition — Both Methods

```
#   Combine both imputation methods for stacked bar
biemer_both <- bind_rows(rq3_biemer_claude, rq3_biemer_mixgb)

biemer_long <- biemer_both |>
  pivot_longer(cols = c(bias_sq, variance),
               names_to = "component",
               values_to = "comp_value") |>
  mutate(
    component = factor(component,
      levels = c("bias_sq", "variance"),
      labels = c("Bias²", "Variance")),
    config = factor(config,
      levels = c("Claude P2", "mixgb S"))
  )

ggplot(biemer_long,
       aes(x = comp_value,
           y = reorder(item, comp_value),
```

```
         fill = component)) +
  geom_col(width = 0.6, position = "stack") +
  geom_text(
    data = biemer_both,
    aes(x = mse,
        y = reorder(item, mse),
        label = paste0(sprintf("%.4f", mse),
                       "  (", pct_syst, "%)"),
        fill = NULL),
    hjust = -0.05, size = 2.5
  ) +
  facet_grid(config ~ survey_label, scales = "free") +
  scale_fill_manual(
    values = c("Bias²"    = viridis(3)[2],
               "Variance" = "grey75"),
    name = NULL
  ) +
  scale_x_continuous(expand = expansion(mult = c(0, 0.40))) +
  labs(
    x = "MSE of Discrimination (a)",
    y = NULL,
    title = "Biemer decomposition: MSE = Bias² + Variance"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    strip.text = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    legend.position = "bottom"
  )
```

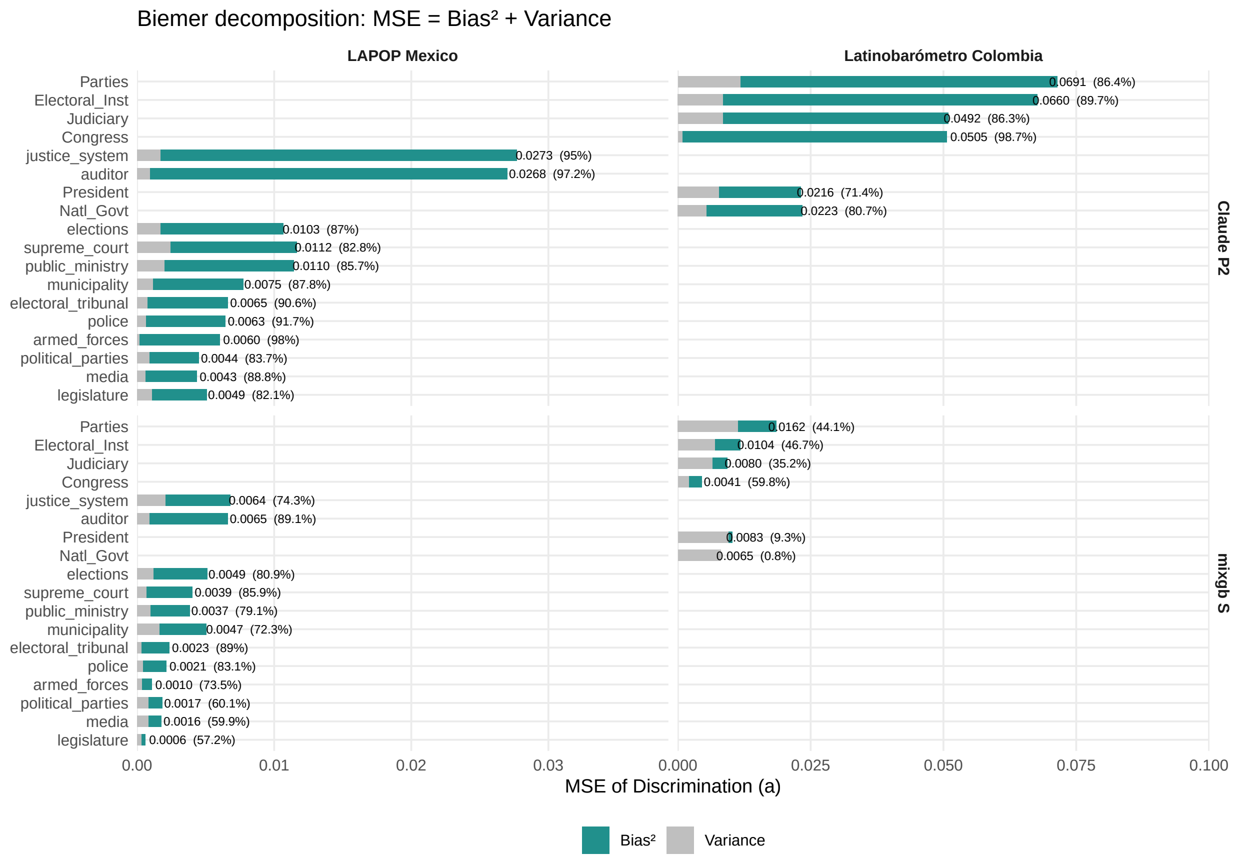

Figure 5: RQ3: Biemer Decomposition of Discrimination MSE — Claude P2 vs mixgb S. Dark fill = Bias² (systematic), light fill = Variance (random). Labels show total MSE and % systematic.

Item-Level Response Variance

```
#   Per-item variance: Gold vs Deletion vs Claude P2
#    vs mixgb S
rq3_item_var <- map_dfr(c("lapop", "lb"), function(s) {
  cfg <- survey_config[[s]]
  items <- cfg$trust_items
  map_dfr(1:5, function(fi) {
    gold_data <- folds_list[[s]] |>
      filter(fold_id == fi) |>
      select(all_of(items)) |>
      mutate(across(everything(), as.numeric))
    del_data <- folds_list[[s]] |>
      filter(fold_id == fi, split == "train") |>
      select(all_of(items)) |>
      mutate(across(everything(), as.numeric))
    claude_data <- reconstruct_full_data(
```

```r
      s, fi, "claude_sonnet", "P2") |>
      select(all_of(items))
    mixgb_recon <- reconstruct_full_data(
      s, fi, "mixgb", "S")
    mixgb_data <- if (!is.null(mixgb_recon)) {
      mixgb_recon |> select(all_of(items))
    } else { NULL }
    map_dfr(items, function(it) {
      tibble(
        survey = s, fold_id = fi, item = it,
        var_gold   = var(gold_data[[it]], na.rm = TRUE),
        var_del    = var(del_data[[it]], na.rm = TRUE),
        var_claude = var(claude_data[[it]], na.rm = TRUE),
        var_mixgb  = if (!is.null(mixgb_data)) {
          var(mixgb_data[[it]], na.rm = TRUE)
        } else { NA_real_ }
      )
    })
  })
}) |>
  mutate(
    survey_label = if_else(survey == "lapop",
      "LAPOP Mexico", "Latinobarómetro Colombia"),
    pct_claude = round((var_claude - var_gold) /
                       var_gold * 100, 2),
    pct_del    = round((var_del - var_gold) /
                       var_gold * 100, 2),
    pct_mixgb  = round((var_mixgb - var_gold) /
                       var_gold * 100, 2)
  )

#   Summary across folds
rq3_var_summary <- rq3_item_var |>
  group_by(survey_label, item) |>
  summarise(
    var_gold   = round(mean(var_gold), 3),
    var_claude = round(mean(var_claude), 3),
    var_del    = round(mean(var_del), 3),
    var_mixgb  = round(mean(var_mixgb, na.rm = TRUE), 3),
    pct_claude = round(mean(pct_claude), 2),
    pct_del    = round(mean(pct_del), 2),
    pct_mixgb  = round(mean(pct_mixgb, na.rm = TRUE), 2),
    .groups = "drop"
  )
```

```r
rq3_var_summary |>
  arrange(survey_label, item) |>
```

```r
  select(Survey = survey_label, Item = item,
         `Gold` = var_gold,
         `Claude` = var_claude, `%C` = pct_claude,
         `mixgb` = var_mixgb, `%M` = pct_mixgb,
         `Del` = var_del, `%D` = pct_del) |>
  knitr::kable(align = "llcccccccc")
```

Table 5: RQ3: Item-Level Response Variance — Gold vs Deletion vs Claude P2 vs mixgb S. pct = % change from gold. Negative = variance shrinkage.

Survey	Item	Gold	Claude	%C	mixgb	%M	Del	%D
LAPOP Mexico	armed_forces	3.310	3.285	-0.78	3.295	-0.45	3.311	0.01
LAPOP Mexico	auditor	2.877	2.877	0.00	2.883	0.20	2.878	0.01
LAPOP Mexico	elections	3.112	3.100	-0.37	3.112	-0.01	3.113	0.01
LAPOP Mexico	electoral_tribunal	3.520	3.514	-0.16	3.521	0.03	3.520	0.00
LAPOP Mexico	justice_system	3.005	2.990	-0.49	2.999	-0.20	3.005	0.00
LAPOP Mexico	legislature	2.998	2.979	-0.63	2.976	-0.73	2.998	0.01
LAPOP Mexico	media	3.339	3.303	-1.08	3.337	-0.06	3.340	0.01
LAPOP Mexico	municipality	2.962	2.942	-0.65	2.956	-0.20	2.961	0.00
LAPOP Mexico	police	3.391	3.388	-0.11	3.394	0.07	3.392	0.01
LAPOP Mexico	political_parties	3.170	3.151	-0.60	3.158	-0.38	3.169	0.00
LAPOP Mexico	public_ministry	3.196	3.170	-0.80	3.180	-0.48	3.195	-0.01
LAPOP Mexico	supreme_court	3.249	3.225	-0.75	3.227	-0.70	3.249	-0.01
Latinobarómetro Colombia	Congress	0.609	0.598	-1.67	0.591	-2.80	0.609	0.01
Latinobarómetro Colombia	Electoral_Inst	0.668	0.661	-0.98	0.653	-2.23	0.668	0.01
Latinobarómetro Colombia	Judiciary	0.690	0.679	-1.50	0.672	-2.57	0.690	0.02
Latinobarómetro Colombia	Natl_Govt	0.824	0.814	-1.31	0.810	-1.79	0.824	0.01
Latinobarómetro Colombia	Parties	0.485	0.485	-0.07	0.476	-1.85	0.485	0.01
Latinobarómetro Colombia	President	1.039	1.011	-2.71	1.025	-1.32	1.039	0.01

Figure: Item Variance Shrinkage

```r
#   Long format for plotting both methods
var_long <- rq3_var_summary |>
  pivot_longer(
    cols = c(pct_claude, pct_mixgb),
    names_to = "method",
    values_to = "pct_change"
```

```r
  ) |>
  mutate(
    method = recode(method,
      pct_claude = "Claude P2",
      pct_mixgb  = "mixgb S"),
    method = factor(method,
      levels = c("Claude P2", "mixgb S"))
  )

ggplot(var_long,
       aes(x = pct_change,
           y = reorder(item, pct_change),
           color = method)) +
  geom_vline(xintercept = 0, linetype = "dashed",
             color = "grey50") +
  geom_point(size = 2.5,
             position = position_dodge(0.4)) +
  facet_wrap(~ survey_label, scales = "free") +
  scale_color_manual(
    values = c("Claude P2" = viridis(4)[3],
               "mixgb S"   = viridis(4)[1]),
    name = NULL
  ) +
  labs(
    x = "% change in item variance (vs Gold)",
    y = NULL,
    title = "Item-level variance shrinkage: Claude P2 vs mixgb S"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    strip.text = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    legend.position = "bottom"
  )
```

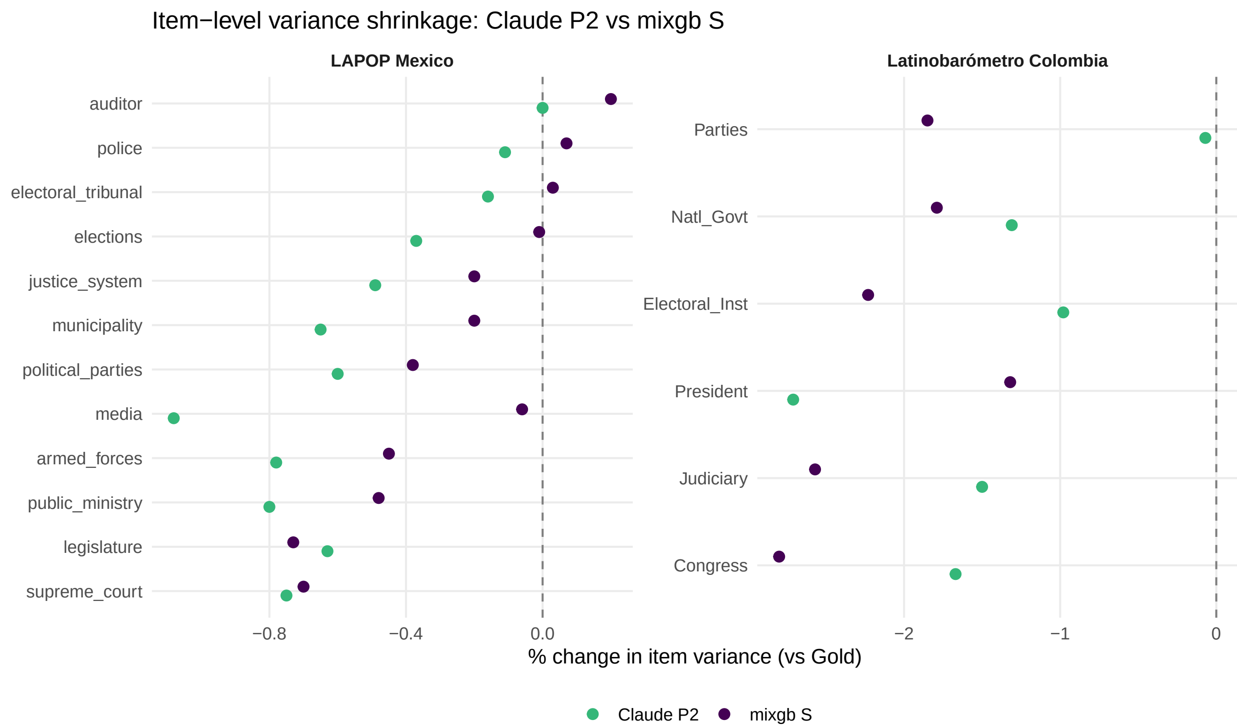

Figure 6: RQ3: Item-Level Variance Change vs Gold. Claude P2 and mixgb S both shrink item response variance; the method with less shrinkage preserves the measurement model's error structure more faithfully.

Save All Results

```r
saveRDS(rq1_item,
  file.path(results_dir, "rq1_item_metrics.rds"))
saveRDS(rq2_person,
  file.path(results_dir, "rq2_person_recovery.rds"))

save(
  # Model fits
  gold_fits, deletion_fits, imputed_fits,
  # RQ1
  rq1_item, rq1_item_summary,
  # RQ2
  rq2_person, rq2_metrics, rq2_summary,
  # RQ3
  rq3_irt_params, rq3_delta_a,
  rq3_biemer_claude, rq3_biemer_mixgb, rq3_biemer_del,
  rq3_item_var, rq3_var_summary,
  file = file.path(results_dir, "evaluation_rq.RData"))

cat("Saved: evaluation_rq.RData,",
    "rq1_item_metrics.rds,",
    "rq2_person_recovery.rds\n")
```

```
Saved: evaluation_rq.RData, rq1_item_metrics.rds, rq2_person_recovery.rds
```

```r
tibble(
  Object = c("gold_fits", "deletion_fits", "imputed_fits",
             "rq1_item", "rq2_person", "rq2_summary",
             "rq3_irt_params", "rq3_biemer_claude",
             "rq3_biemer_mixgb", "rq3_item_var"),
  Rows = c(nrow(gold_fits), nrow(deletion_fits),
           nrow(imputed_fits), nrow(rq1_item),
           nrow(rq2_person), nrow(rq2_summary),
           nrow(rq3_irt_params), nrow(rq3_biemer_claude),
           nrow(rq3_biemer_mixgb), nrow(rq3_item_var))
) |> knitr::kable()
```

Object	Rows
gold_fits	10
deletion_fits	10
imputed_fits	120
rq1_item	120
rq2_person	30036

Object	Rows
rq2_summary	24
rq3_irt_params	2160
rq3_biemer_claude	18
rq3_biemer_mixgb	18
rq3_item_var	90